Calcurion Maths Worksheets

A-Level Mathematics

Topic: Mechanics – Forces & Newton's Laws

MARKSCHEME ($g = 9.8\,\text{m s}^{-2}$)

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1. $a = \frac{F}{m} = 5$ **M1A1** [2]
2. $F = ma = 15$ **M1A1** [2]
3. $a = 3$ **M1A1** [2]
4. $a = 2$; $s = \frac{1}{2} \times 2 \times 16 = 16$ **M1A1** [5]
5. Resolve: $20 - 2g\sin 30° = 2a \Rightarrow a = 5.1$ **M1A1** [5]
6. Comp. $= 4g\sin 20°$; $a = g\sin 20°$ **M1A1** [4]
7. $R = \sqrt{25 + 144} = 13$ **M1A1** [3]
8. Resultant $5\mathbf{i} + \mathbf{j}$; $a = \mathbf{i} + 0.2\mathbf{j}$ **M1A1** [4]
9. $a = 4$; $v = 12$, $s = 18$ **M1A1** [4]
10. $T - 800g = 800 \times 1.5 \Rightarrow T = 9040$ **M1A1** [4]
11. $600g - T = 600 \times 0.5 \Rightarrow T = 5580$ **M1A1** [4]
12. $F = 3g\sin 40°$ **M1A1** [4]
13. $10\sqrt{3}$, 10 **M1A1** [3]
14. $a = 13$ **M1A1** [3]
15. $\mathbf{F} = -2\mathbf{i} - 5\mathbf{j}$ **M1A1** [3]
16. $a = \frac{40 - 5g\sin 30°}{5} = 3.1$; $s = \frac{1}{2}at^2$ **M1A1** [6]
17. $a = -g\sin 25°$; $0 = 64 + 2as \Rightarrow s = \frac{64}{2g\sin 25°}$ **M1A1** [6]
18. $30 - F = 5 \times 2 \Rightarrow F = 10$ **M1A1** [4]
19. Resultant $5\mathbf{i} + 3\mathbf{j}$; $|F| = \sqrt{34}$; $a = \frac{1}{5}(5\mathbf{i} + 3\mathbf{j})$ **M1A1** [5]
20. $T = 570 \times 11.8$; $R = 70 \times 11.8$ **M1A1** [6]
21. $a = \frac{25 - 3g\sin 35°}{3}$ **M1A1** [5]
22. $a = 4\mathbf{i} + \mathbf{j}$; position $8\mathbf{i} + 2\mathbf{j}$ **M1A1** [6]
23. $\mathbf{F} = -\mathbf{i} - 4\mathbf{j}$ **M1A1** [3]
24. $F - 200 = 1200 \times 2 \Rightarrow F = 2600$ **M1A1** [5]
25. $mg\sin\alpha = ma \Rightarrow a = g\sin\alpha$ **M1A1** [3]